

Simultaneous-choice calibration readout

EXPERIMENTAL - RETAINED STARTING POINT | Experimental simultaneous tenure and birth-attempt choice

The historical runs reproduce exactly, but the experimental fit remains poor. The 32-start search returned 27 valid histories, three market-convergence failures and two timeouts, without improving the retained starting point. All eleven parameters were free; no differential-evolution or polishing round fit the time budget. Twenty-one local sensitivity histories and two exact repetitions then completed. This is a bounded search, not a fully optimized replacement calibration.

Selected objective: 450.705. All twelve target rows and eleven free parameters appear in the following tables.

Retained benchmark objective: 30.483. The target and weight rows match exactly. Its full fit and parameter tables appear in the appendix; the shock specification differs.

Reproduction: Two final histories reproduce the selected fit and all seventeen standard graphs exactly.

Policy status: PARTIAL policy evidence: 33 certified full-branch dates plus 7 valid property-tax prefix dates, 40 total, 4 unavailable after the property-tax branch fails market clearing in 2051. No overall pass is asserted.

Interpretation: The parent/nonparent ownership gap is the main problem: 0.44 percentage points in the model versus 16.77 points in the target, contributing 379.21 of the 450.71 objective. First-birth rooms rise by 0.365 versus the 0.720 target, and births occur too early. Smaller-shock starts improve ownership differences but worsen fertility fit. Other parameters also vary, so this is not an optimized trade-off or proof that simultaneous choice cannot fit the data.

Outstanding issues: Resolve market clearing at the finite-grid affordability threshold and review the nesting restriction before another large search. The tested tenure nests require fertility taste dispersion to be no larger than tenure dispersion; simultaneous realization alone does not impose that restriction. All targets should be retained. The first-child room-floor jump has an upper bound of 2 rooms, versus 0.5 in the benchmark; high-income wealth reaches the upper plotted boundary. Production remains the presentation benchmark. Low-wealth conditional policies remain irregular despite passed occupied-state checks.

Economic specification. Tenure and birth attempts are chosen jointly after the taste shocks are observed. Tenure is committed before conception success; housing size, consumption and saving can adapt within tenure afterward. Products within tenure are chosen deterministically. The inner taste scale is λk , with $0 < \lambda \leq 1$. The experiment changes shock structure as well as timing.

Policy closure. Each date clears housing markets with current prices treated as permanent. After 2023, outside entry is zero, retention is one, and the inherited four-slot birth queue converts births to new households at 1/2.1. Supply elasticity is fixed at 0.63. Tax revenue is discarded. These are finite paths under temporary equilibrium.

Complete target fit

Shares and share gaps are fractions: 0.01 equals one percentage point. The objective is the sum of $\text{weight} \times \text{gap}^2$; weights need not equal empirical precision.

Moment	Target	Model	Gap	Weight	Loss contribution
Completed fertility	1.918	1.91334	-0.00466471	1425.74	0.0310234
Childlessness	0.188	0.19116	0.0031602	17180.7	0.171582
Mean age at first birth	26.0446	25.7564	-0.288232	44.4444	3.69235
Share first births age 30+	0.260327	0.204587	-0.0557406	10000	31.0702
Rooms response to first birth	0.720246	0.365484	-0.354762	137.565	17.3134
Rooms: 3+ minus 1–2 children	0.3677	0.363062	-0.00463792	2958.51	0.0636387
Parent/nonparent ownership gap, ages 30-55	0.167662	0.00441631	-0.163245	14229.6	379.205
Ownership, ages 30-55	0.575472	0.501645	-0.0738265	1207.85	6.58319
Mean occupied rooms	5.77997	5.32431	-0.455659	11.9732	2.48592
Wealth / annual earnings	6.8731	7.16305	0.28995	6.28767	0.528609
Bequest flow / wealth	0.0088	0.00825427	-0.000545731	5.16529e+06	1.53834
Wealth/income p90/p50, ages 76-84	3.44811	3.07283	-0.375281	56.9598	8.02199

Full-table objective check: sum of contributions = 450.705293146.

For the ownership gap, the active model parent group has at least one child at home; the nonparent group has no previous birth. The code retains a legacy new-parent variable name.

Free parameters, bounds, and fixed/derived entries

Free parameter	Value	Lower	Upper	Near bound?
Annual discount factor	0.996308	0.94	0.9995	False
Outer taste scale	2.03837	0.005	10	False
Nest dissimilarity	0.995122	0.02	1	True
Owner housing-service premium	1.25167	0.1	5	False
Housing-supply normalization	11.5047	0.2	80	False
Bequest utility weight	0.595489	0	8	False
Bequest wealth shift	0.092875	0.02	16	True
Per-child room floor	0.293129	0.1	1.8	False
First-birth utility cost	4.92	0	8	False
First-child room-floor jump	0.395445	0	2	False
Child-value change, 2007-23	-0.378933	-1.5	0.2	False

Near-bound flags use the existing convention: within 2% of the full physical parameter range, rather than distance in the transformed search coordinate.

Fixed/derived entry	Value	Recorded status
Child value, 2007	0.258572	Normalized to old completed fertility 2.1
Child value, 2023	-0.120362	Old level plus fitted 2007–23 change
Housing-supply elasticity	0.63	Externally fixed

The 12-by-11 weighted local sensitivity matrix has numerical rank 11 and condition number about 1,271 in transformed parameter coordinates. The housing-supply normalization uses a one-sided difference because its negative perturbation failed market clearing. The weakest local direction is dominated by the bequest wealth-shift parameter. These finite differences do not establish global identification. Several unselected probes improve the objective slightly, confirming that the retained point is not a local optimum.

Policy outcomes and verification coverage

Policy	Year	Births (%)	Ownership (pp)	Rooms (%)
Housing supply +20%	2023	0.929532	0.0650921	6.20088
Housing supply +20%	2063	2.8287	0.0172638	7.17689
Dependent-child LTV 95%	2023	0.519493	0.0205751	0.409257
Dependent-child LTV 95%	2063	1.84469	0.00878045	0.0287035
Property tax doubled	2023	-1.05778	0.00728305	-4.86911
Property tax doubled	2047	-1.61731	-0.00302925	-5.42804

Each effect compares the policy with the baseline at the same date and inherited starting population. The table shows each policy's first and last available date; complete dated results are retained in its recorded path file.

PARTIAL policy evidence: 33 certified full-branch dates plus 7 valid property-tax prefix dates, 40 total, 4 unavailable after the property-tax branch fails market clearing in 2051. No overall pass is asserted.

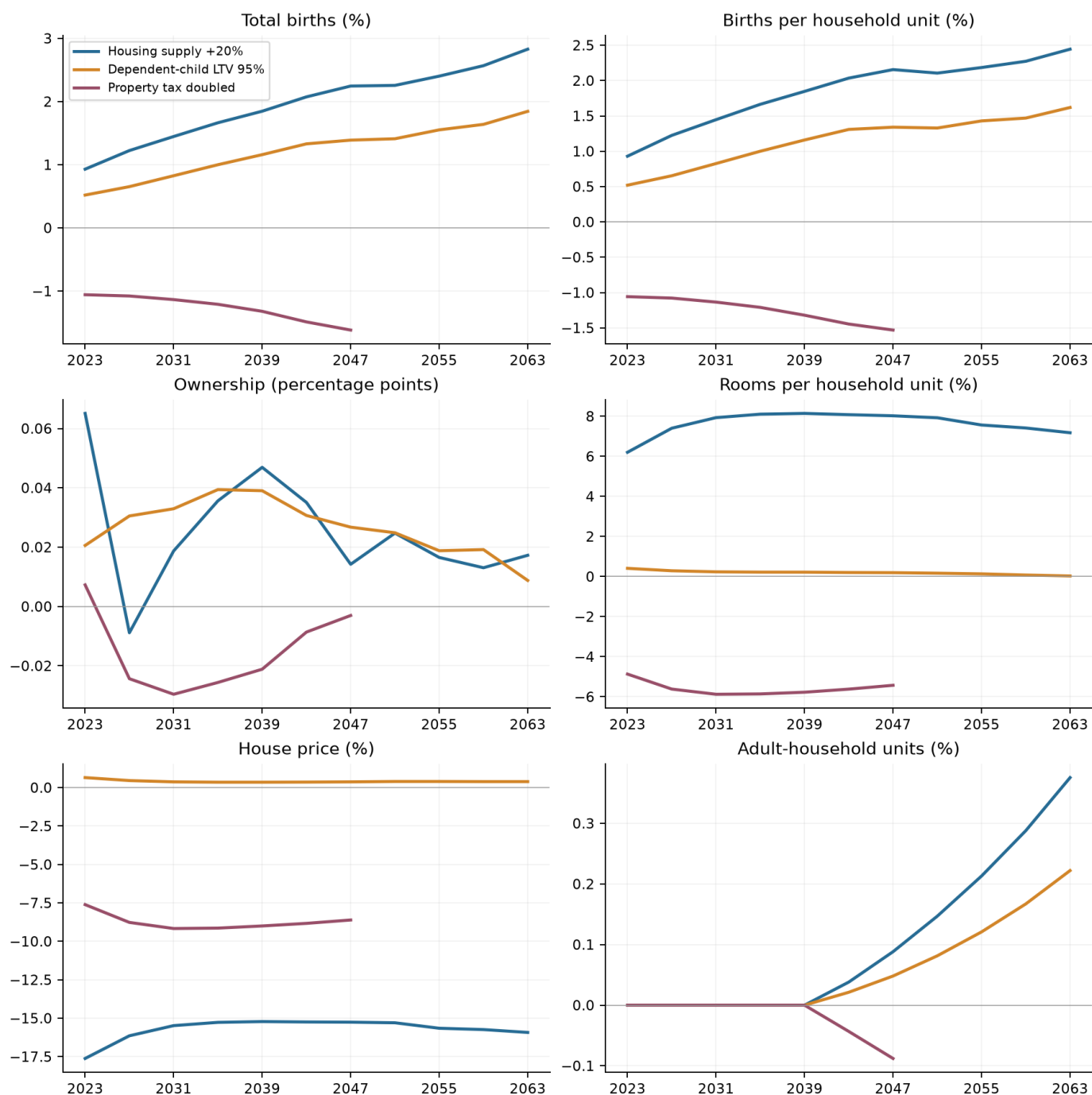
All 40 dated packets contain the unchanged seventeen graphs and pass the occupied-value, budget and probability checks. Maximum budget-violating mass: 0, below $2e-10$.

At this experimental vector, additional supply raises births by 0.93% initially and 2.83% in 2063; the mortgage policy raises them by 0.52% and 1.84%, with very small ownership changes. The large historical ownership miss limits these policy interpretations. Baseline adult-household mass falls to 71.18% of its 2023 level under the closed-entry experiment; this is not an empirical population forecast.

The tax branch fails at 2051, with residual 0.0002791 against the unchanged 0.0002 gate. A 72-price fixed-state trace finds a demand jump across prices only $8.19e-11$ apart. Exact endpoint replays identify occupied renters switching from four to two rooms at the down-payment threshold. For wealth 0.4418605 and an 80% financed share, a four-room home becomes unaffordable above price $0.4418605 / (0.2 \times 4) = 0.5523256$. Both endpoints pass budget and value checks. This supports a finite-grid affordability discontinuity; it does not complete the tax path or establish economic nonexistence.

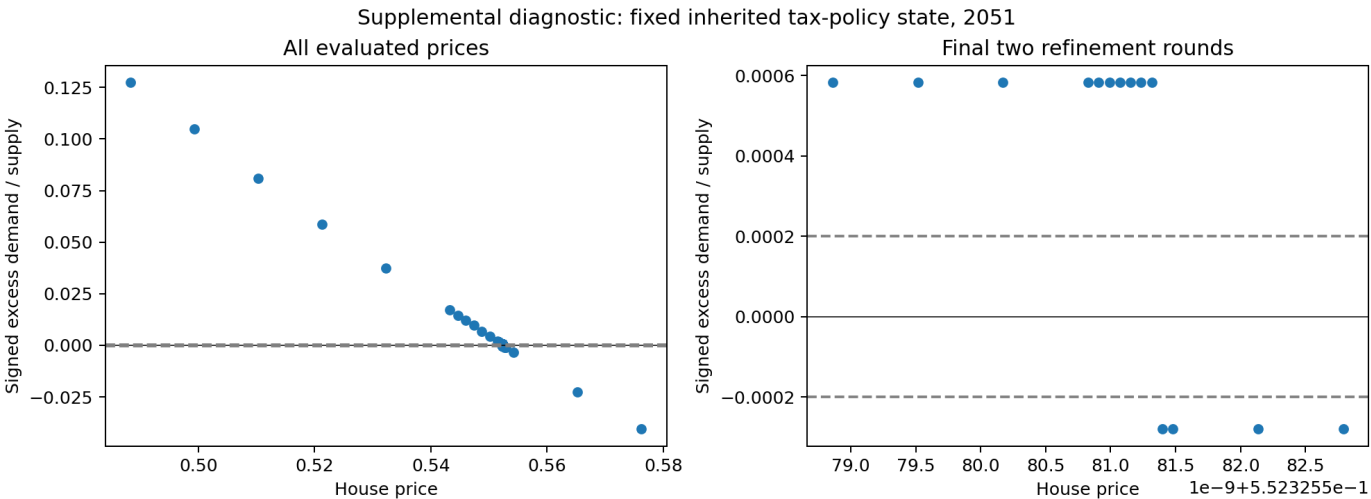
Supplemental: policy paths

Effects relative to the baseline at each date. Total births include changes in adult-household units; births per household unit isolate the flow per unit. These units are not resident-person counts. Ownership is a percentage-point difference; the other panels show percent changes. The paths use temporary equilibrium and the experimental vector.



Supplemental: market-clearing diagnostic

This price trace holds the inherited 2051 tax-policy population and parameters fixed. It diagnoses the rejected market solve and does not complete or replace the policy path.



Retained benchmark: complete target fit

Moment	Target	Model	Gap	Weight	Loss contribution
Completed fertility	1.918	1.92291	0.0049073	1425.74	0.034334
Childlessness	0.188	0.189407	0.00140702	17180.7	0.034013
Mean age at first birth	26.0446	26.256	0.211404	44.4444	1.9863
Share first births age 30+	0.260327	0.237579	-0.0227482	10000	5.17479
Rooms response to first birth	0.720246	0.436418	-0.283828	137.565	11.0821
Rooms: 3+ minus 1–2 children	0.3677	0.403441	0.0357414	2958.51	3.77934
Parent/nonparent ownership gap, ages 30-55	0.167662	0.161699	-0.00596315	14229.6	0.505992
Ownership, ages 30-55	0.575472	0.544488	-0.0309842	1207.85	1.15956
Mean occupied rooms	5.77997	6.31729	0.537321	11.9732	3.45681
Wealth / annual earnings	6.8731	6.93265	0.0595522	6.28767	0.022299
Bequest flow / wealth	0.0088	0.00843262	-0.000367376	5.16529e+06	0.697133
Wealth/income p90/p50, ages 76-84	3.44811	3.23651	-0.2116	56.9598	2.55034

Complete-table objective: 30.482966708. This is the retained production comparison; experimental search does not promote a replacement.

Retained benchmark: all parameters and restrictions

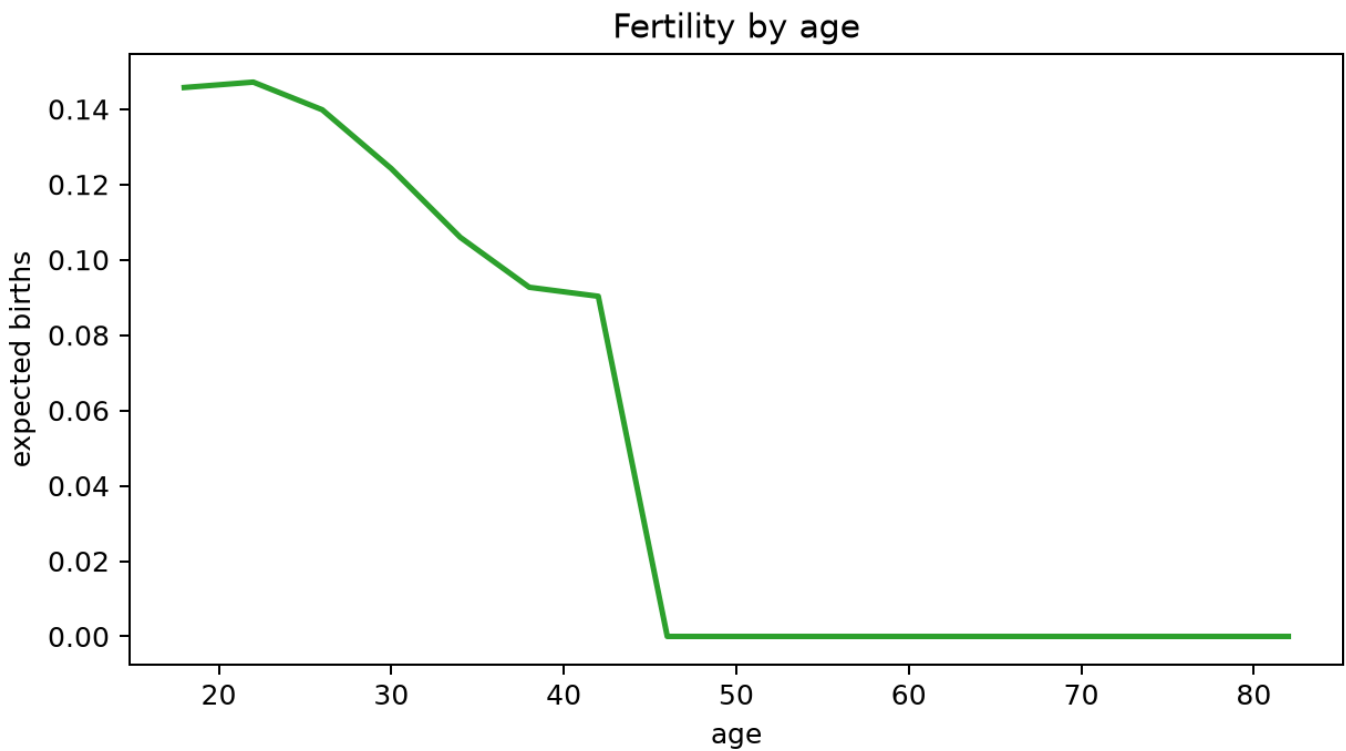
Parameter	Value	Lower	Upper	Free?	Near bound?
Annual discount factor	0.995117	0.94	0.9995	True	False
First-birth taste dispersion	2.16817	0.02	50	True	False
Later-birth taste dispersion	1.73647	0.02	50	True	False
Owner housing-service premium	1.04347	0.1	5	True	False
Housing-supply normalization	14.563	0.2	80	True	False
Bequest utility weight	0.528428	0	8	True	False
Bequest wealth shift	0.107249	0.02	16	True	True
Per-child room floor	0.28221	0.1	1.8	True	False
First-birth utility cost	4.55914	0	8	True	False
First-child room-floor jump	0.364931	0	0.5	True	False
Child-value change, 2007-23	-0.328714	-1.5	0.2	True	False
Child value, 2007	0.288017	-	-	False	False
Child value, 2023	-0.0406971	-	-	False	False
Tenure taste dispersion	0.005	-	-	False	False
Housing-supply elasticity	0.63	-	-	False	False

The old child-value intercept is normalized to completed fertility 2.1; its 2023 level follows from the estimated change. Housing-supply elasticity and the tenure taste scale are externally fixed. The original model estimates separate first- and later-birth taste scales.

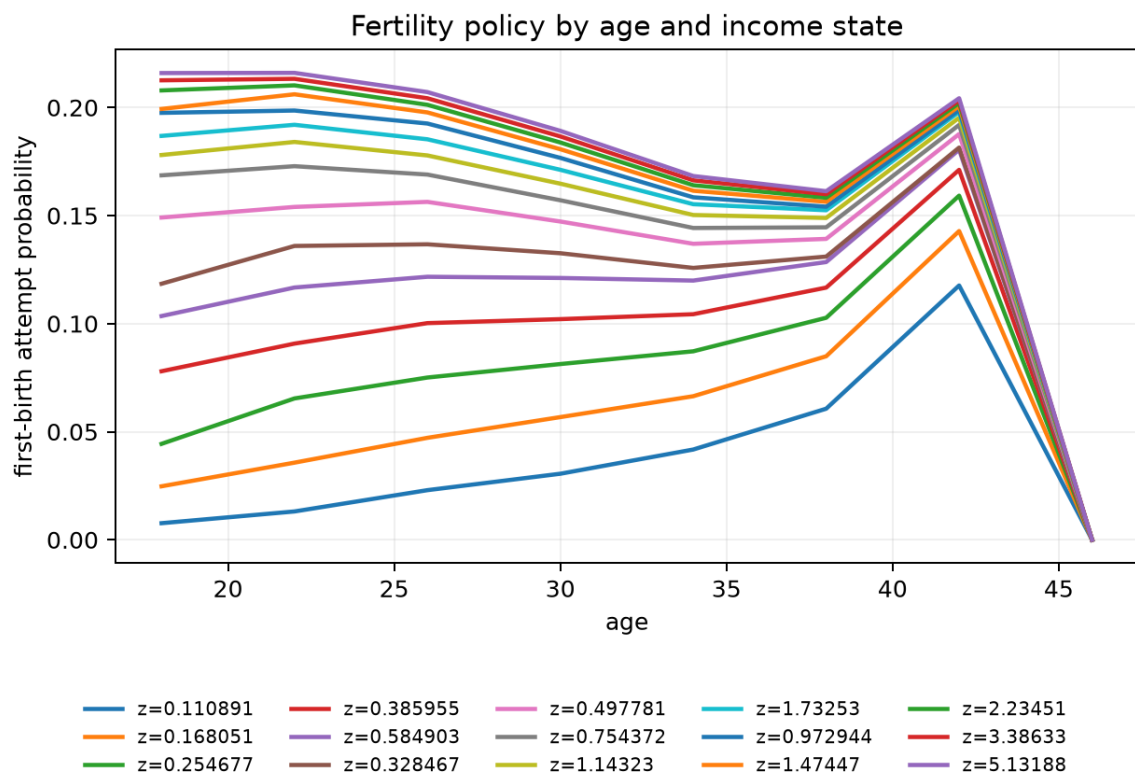
Standard diagnostic graphs

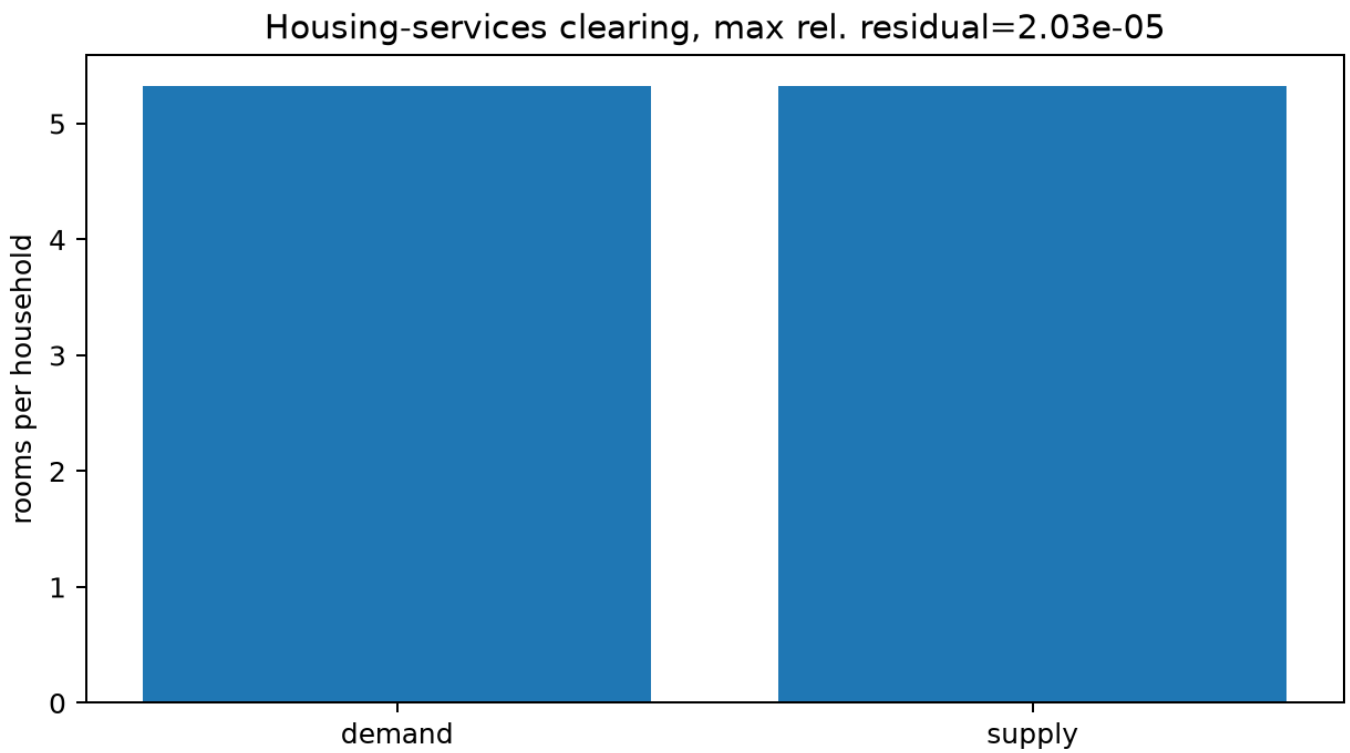
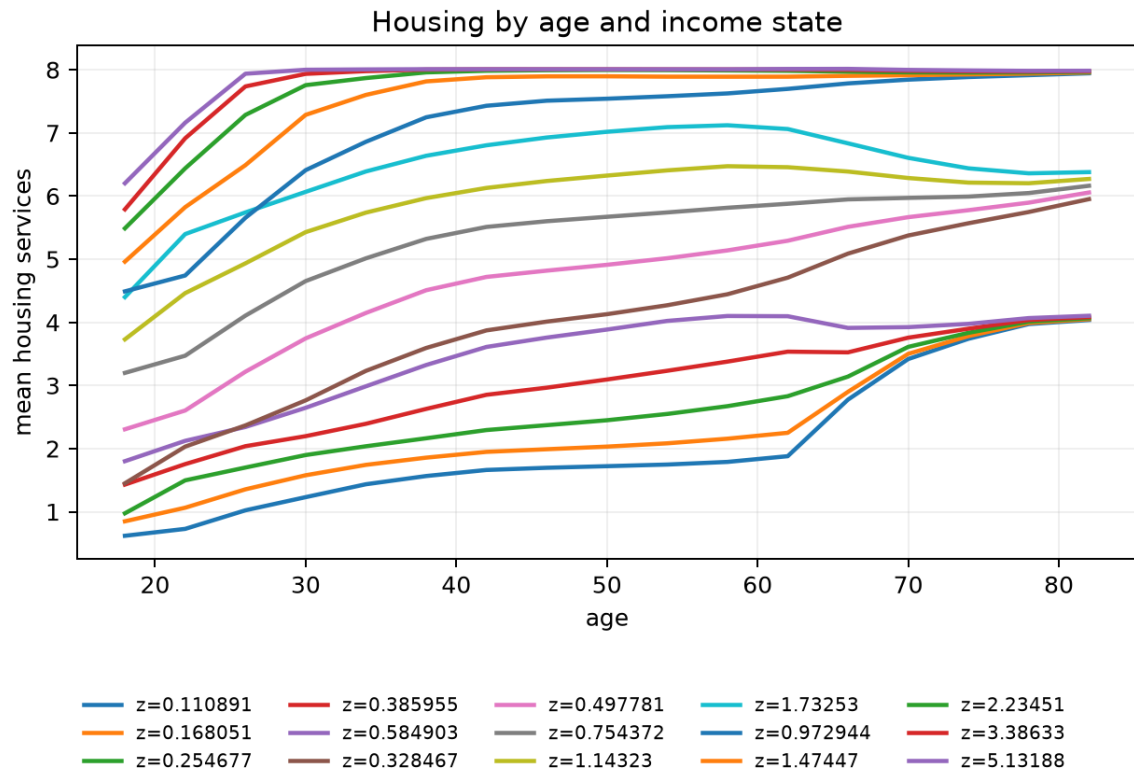
The established seventeen-graph set and its data are preserved. Crowded legends and axis labels have been repositioned for legibility; the original files remain in the verified source packet. These are selected-candidate diagnostics.

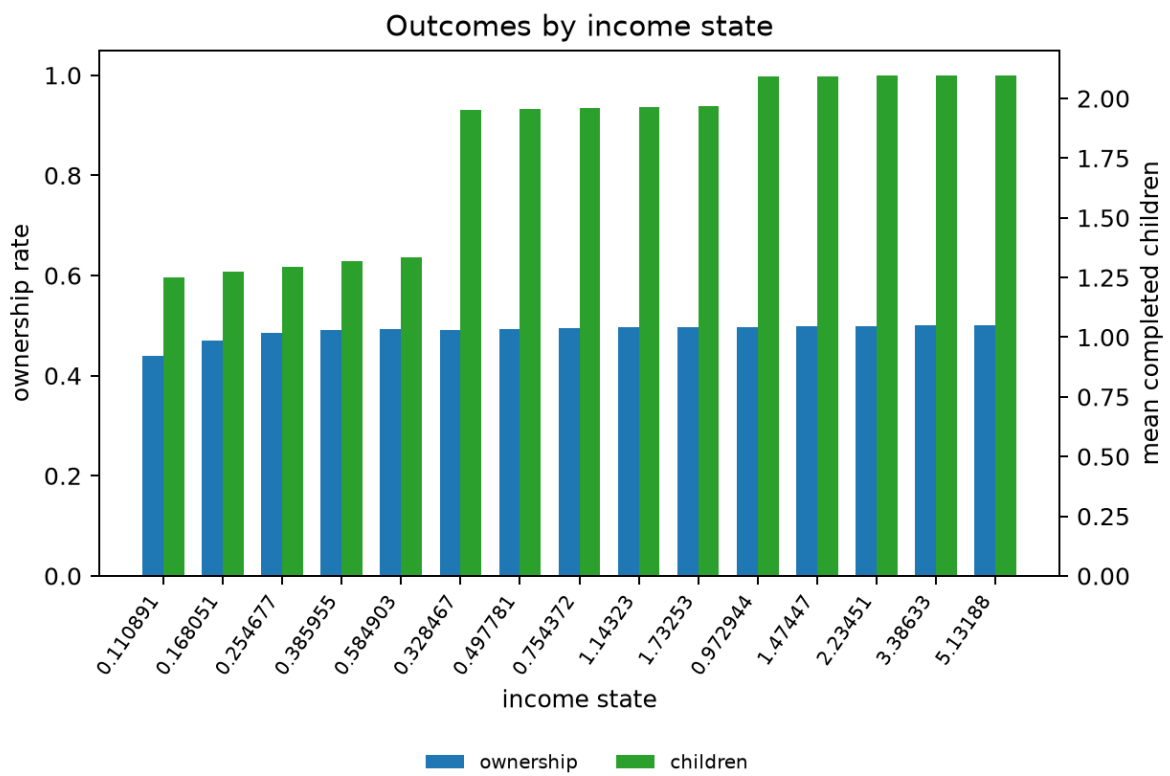
fertility by age

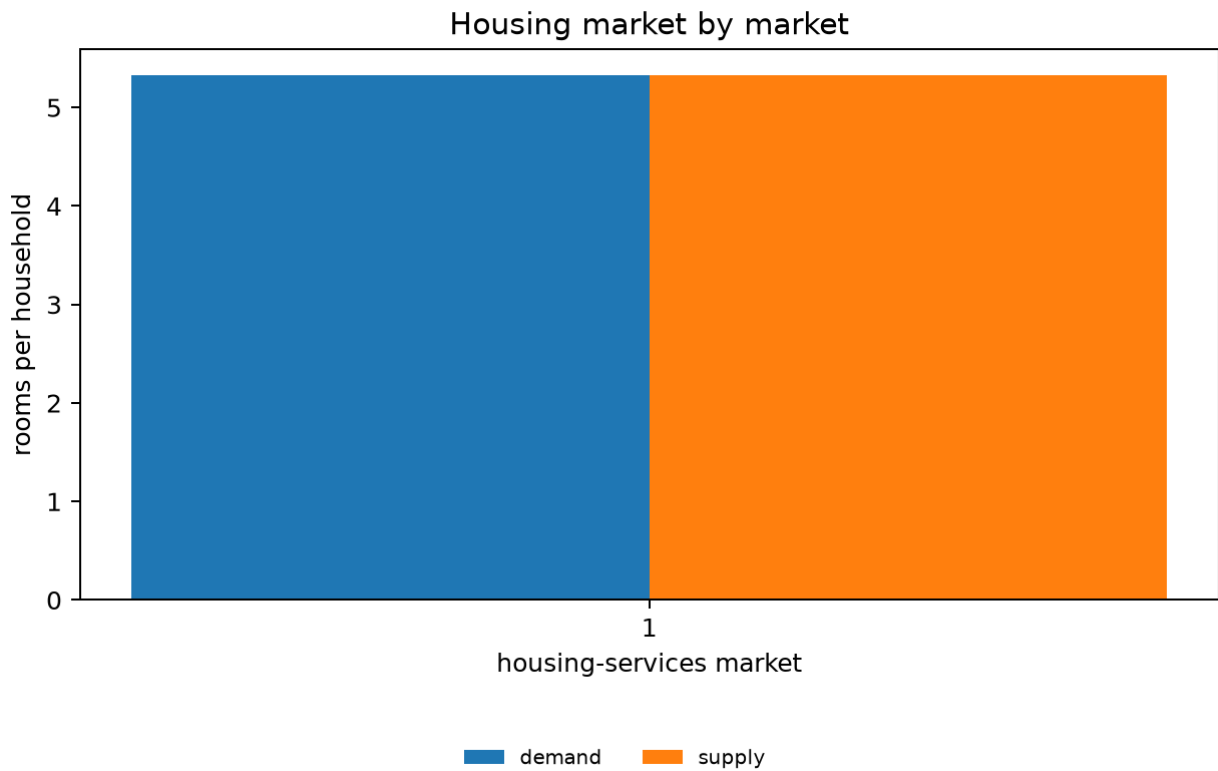
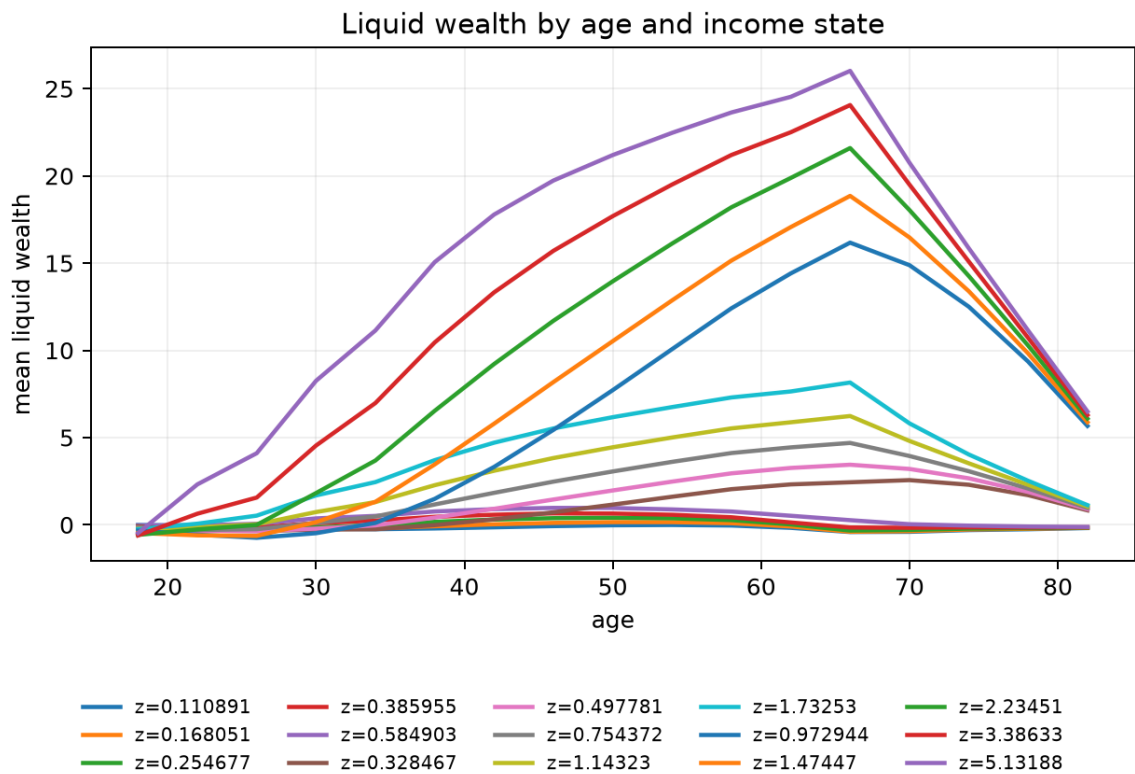


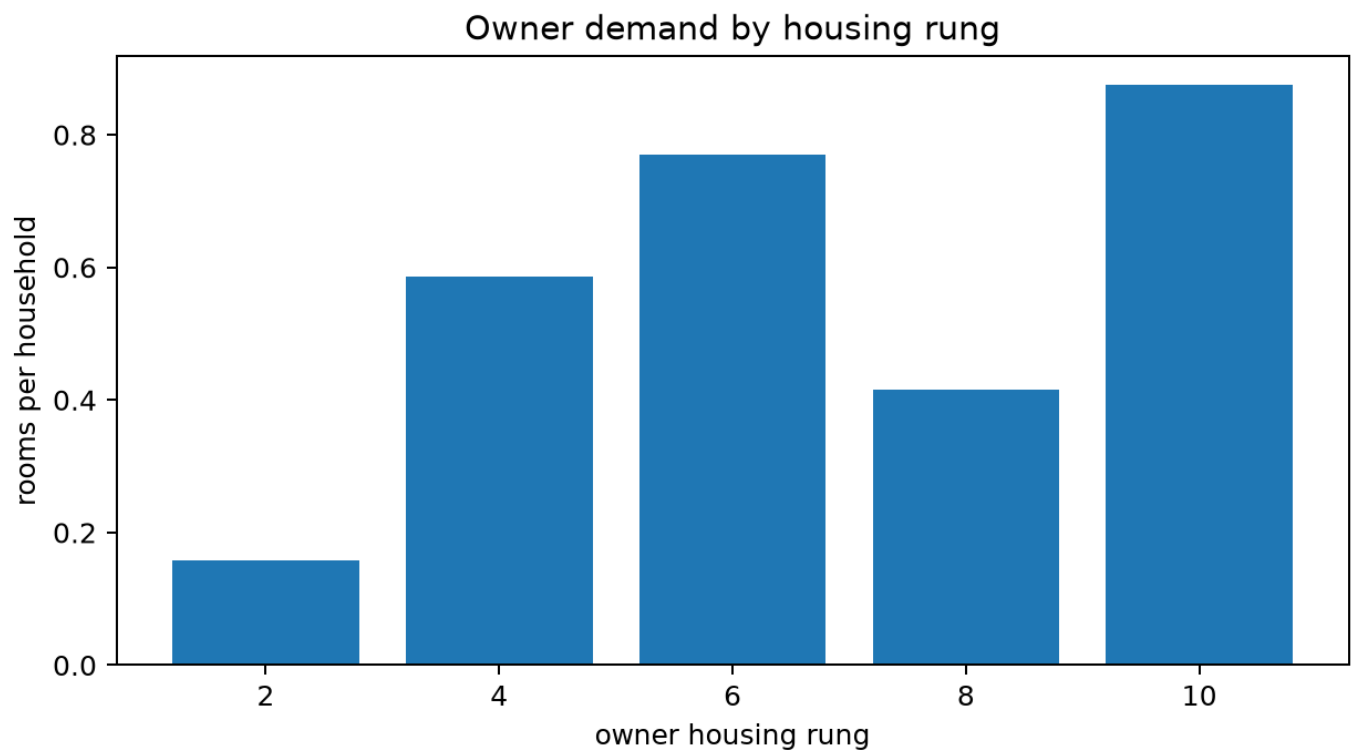
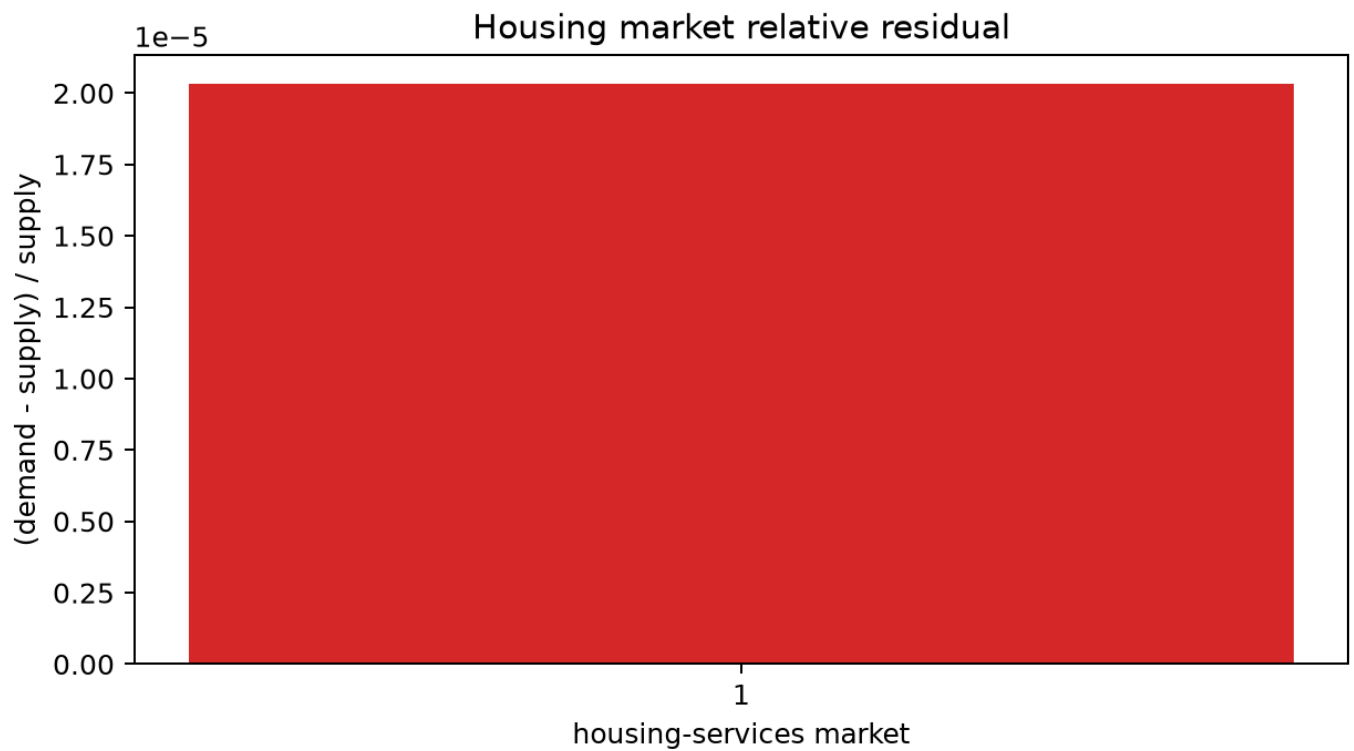
fertility policy by age income state

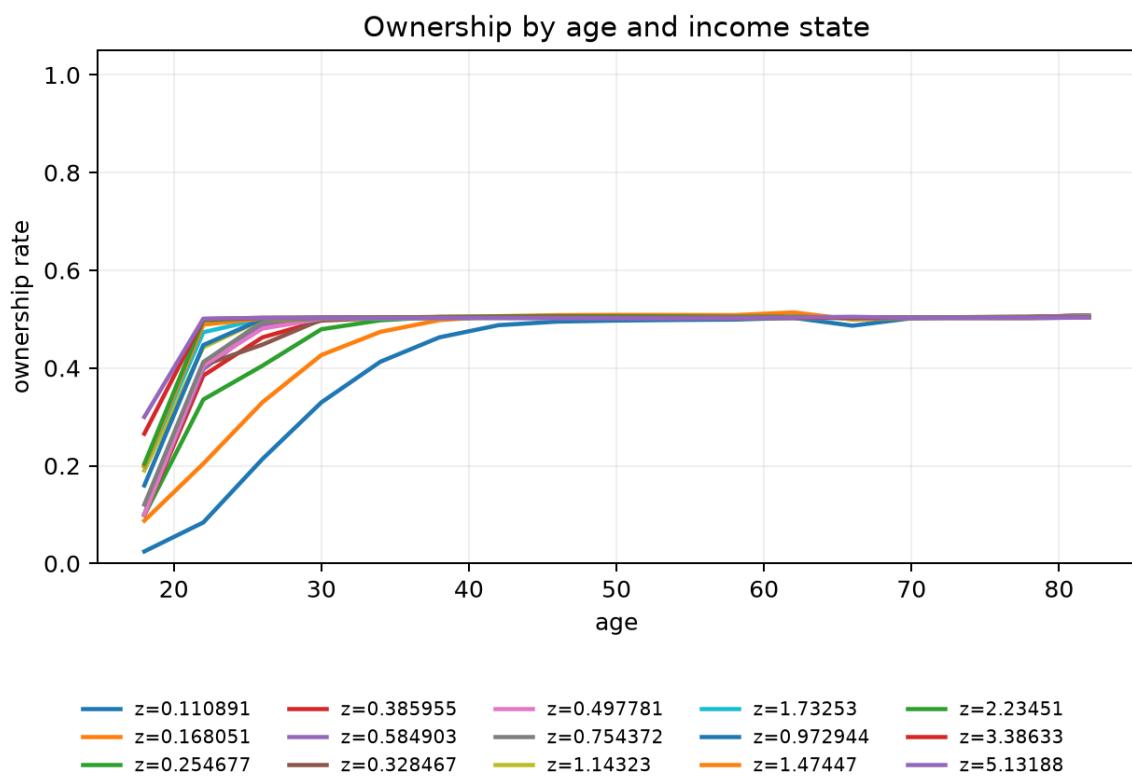
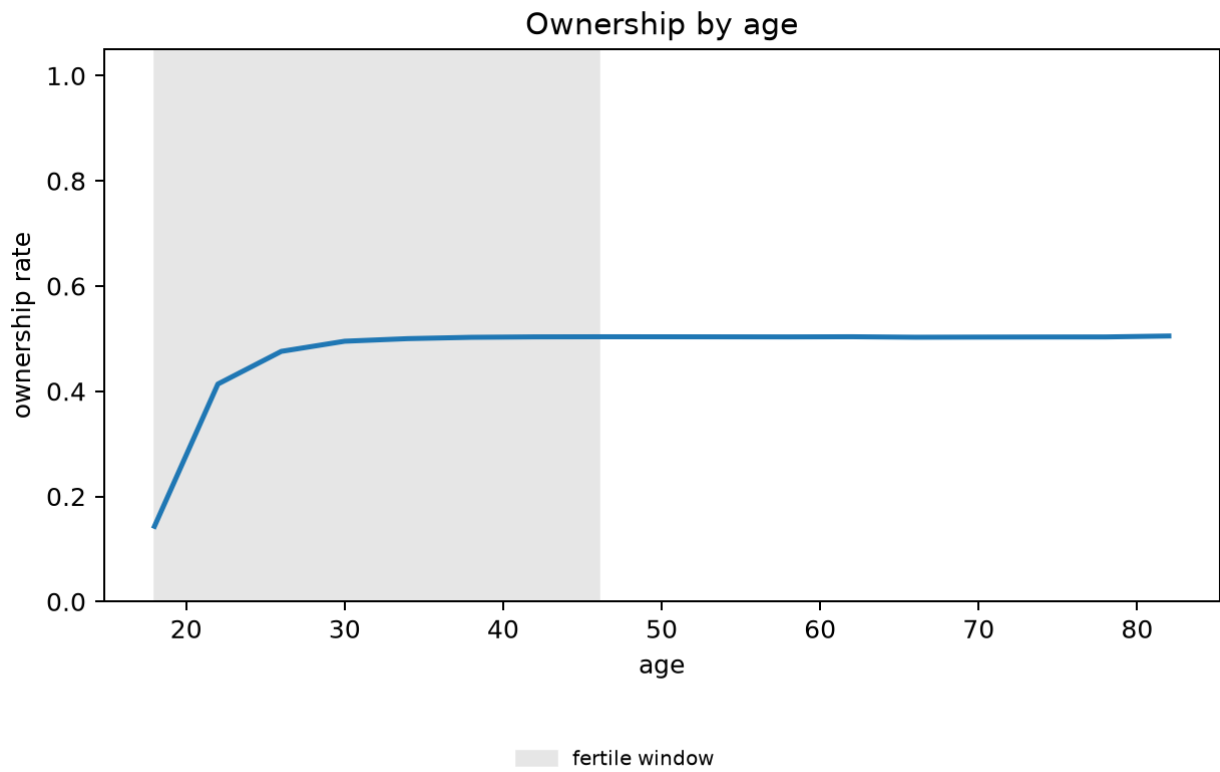




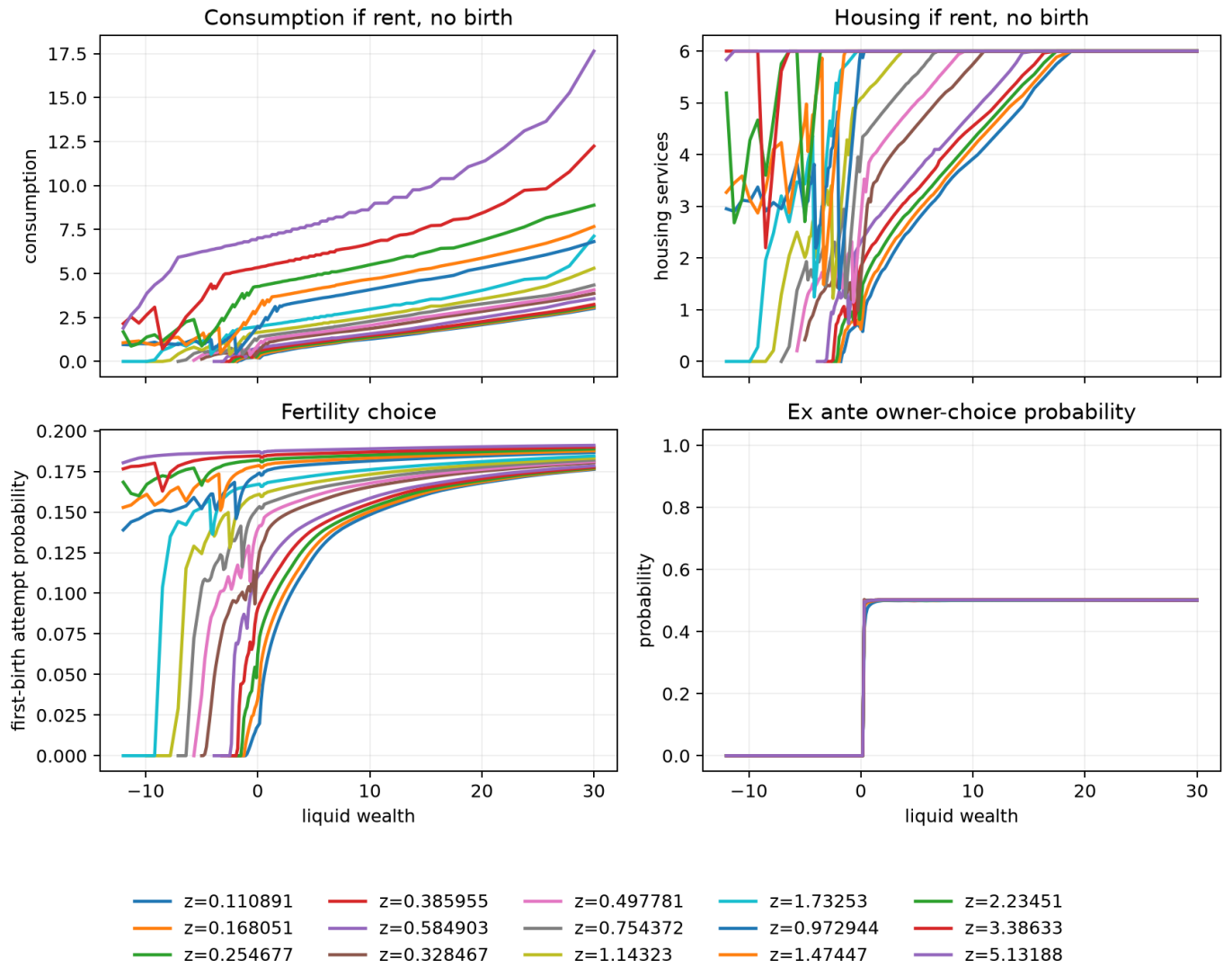




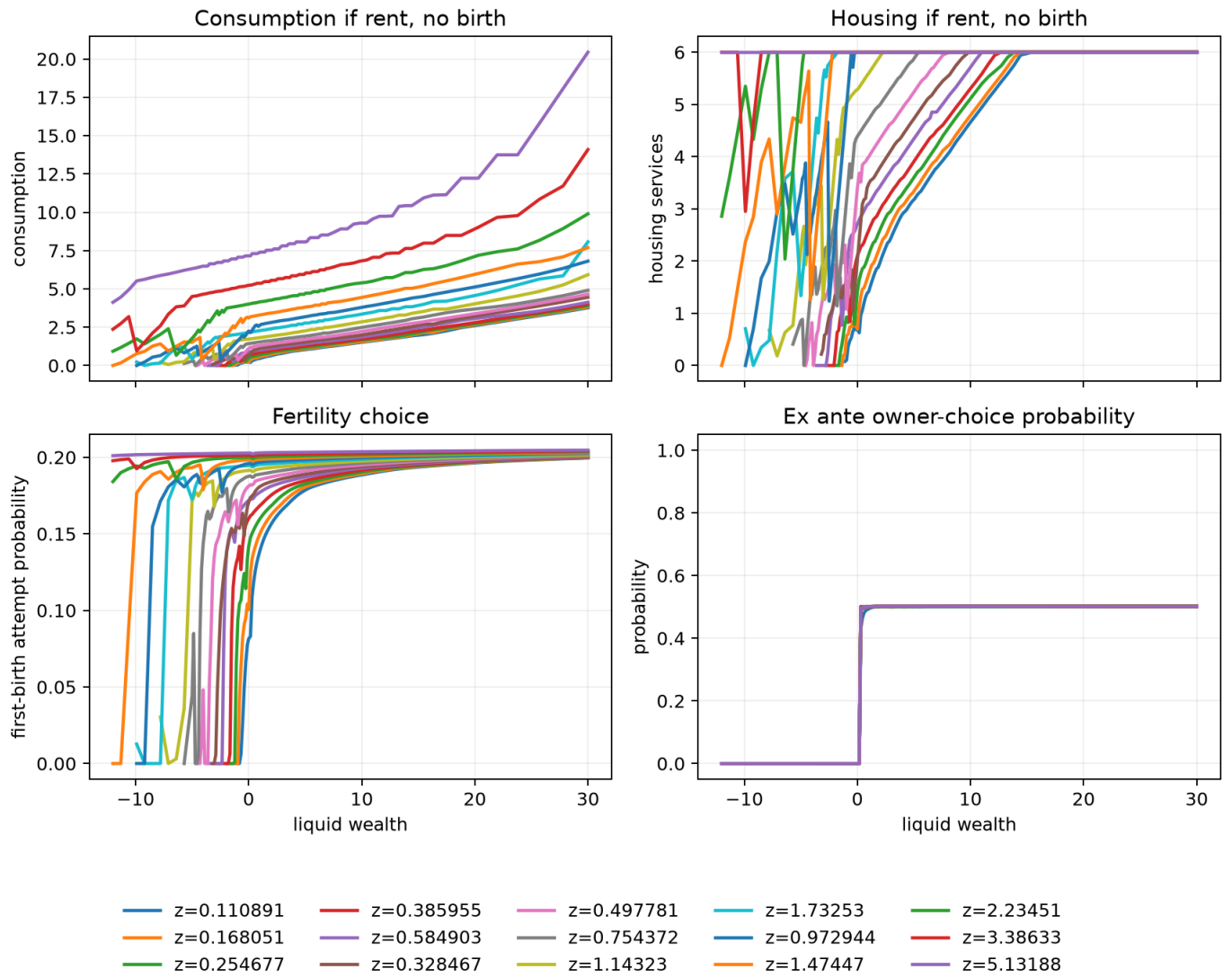


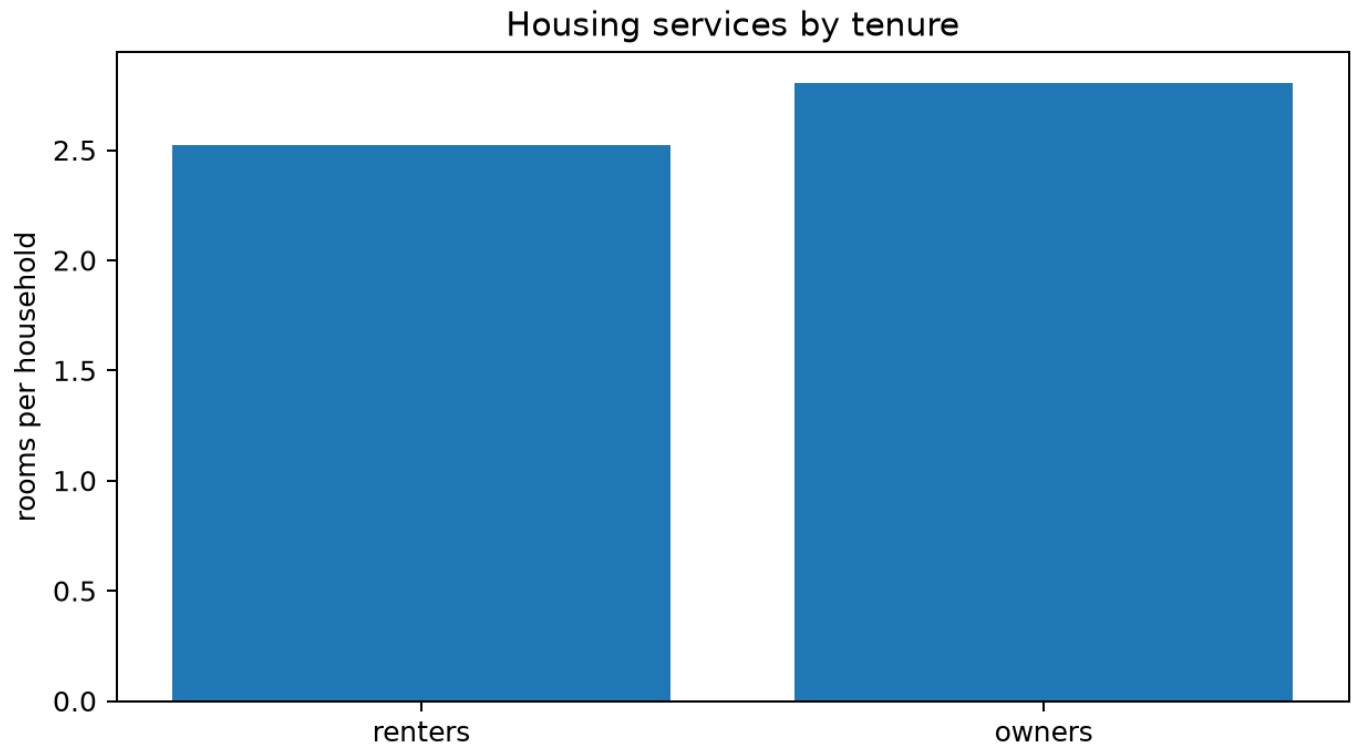


Childless renter policies at age 30

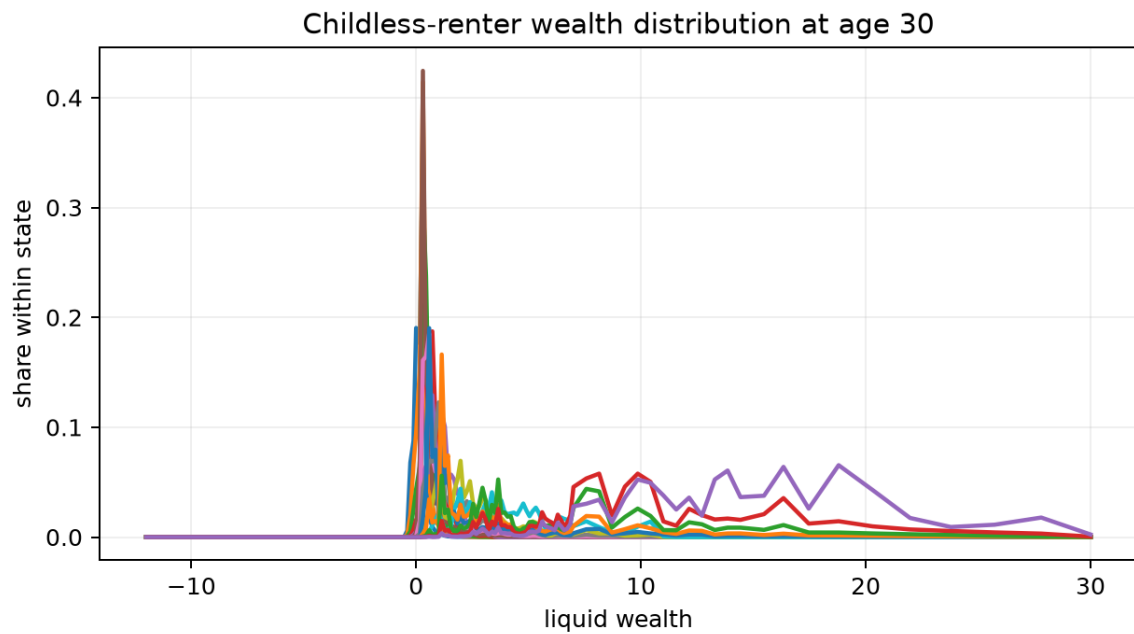


Childless renter policies at age 42





wealth dist childless renter age30



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z=0.254677	z=0.328467	z=1.14323	z=1.47447	z=5.13188

